AIR QUALITY PREDICTION USING MACHINE LEARNING IN R

A Time-Aware Comparative Study of Regression Models

Course: Predictive Analytics (CSE574)
Language/IDE: R / RStudio

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Faculty: _____
Submission Date: _____

Abstract

This project estimates same-hour carbon monoxide concentration (CO(GT), mg/m³) from sensor responses, weather and calendar features. It uses the UCI Air Quality dataset, handles -200 as missing, and compares five regression algorithms with a chronological train/validation/test split. MAE, RMSE and R^2 evaluate the held-out later period. A high-pollution error audit adds a robustness check. This is same-hour estimation, not a future forecast.

1. Problem Statement and Objectives

Research question: how accurately can same-hour CO concentration be estimated from sensor-response, weather and calendar features, and which regression algorithm performs best on a later held-out period?

- Clean and validate the source data.
- Explore CO distribution and temporal patterns.
- Compare Multiple Linear Regression, KNN, linear SVR, Random Forest and Gradient Boosting.
- Use time-aware validation and training-only imputation/scaling.
- Report test metrics and inspect higher-CO errors.

2. Dataset and Feature Design

Source: UCI Air Quality dataset (hourly observations from an Italian city in 2004–2005). The CSV uses semicolons and decimal commas and includes trailing empty records/columns. The target is CO(GT) in mg/m³. Predictors are five PT08 sensor responses, T, RH, AH, hour, month and day of week. NMHC(GT) is excluded due to extreme missingness; other reference-analyzer pollutants are excluded to keep the experiment sensor/weather-oriented.

3. Novelty and Scope

The practical novelty is a sensor-and-weather-only feature set, chronological evaluation, five-model comparison, and high-pollution error audit. It is a same-hour estimation task—not a next-hour or next-day forecast.

4. Methodology

- Remove fully blank rows/trailing empty columns; convert -200 to NA; parse timestamps; retain rows with valid CO target.
- Split chronologically: first 70% train, next 15% validation, final 15% test.
- Fit median imputation using training data only; scale KNN/SVR using training means and standard deviations.
- Tune hyperparameters on validation RMSE; refit selected settings on train+validation; evaluate once on untouched test data.
- Metrics: MAE and RMSE in mg/m³; R^2 for explained variation. Classification accuracy is not appropriate for a continuous target.

5. Exploratory Data Analysis

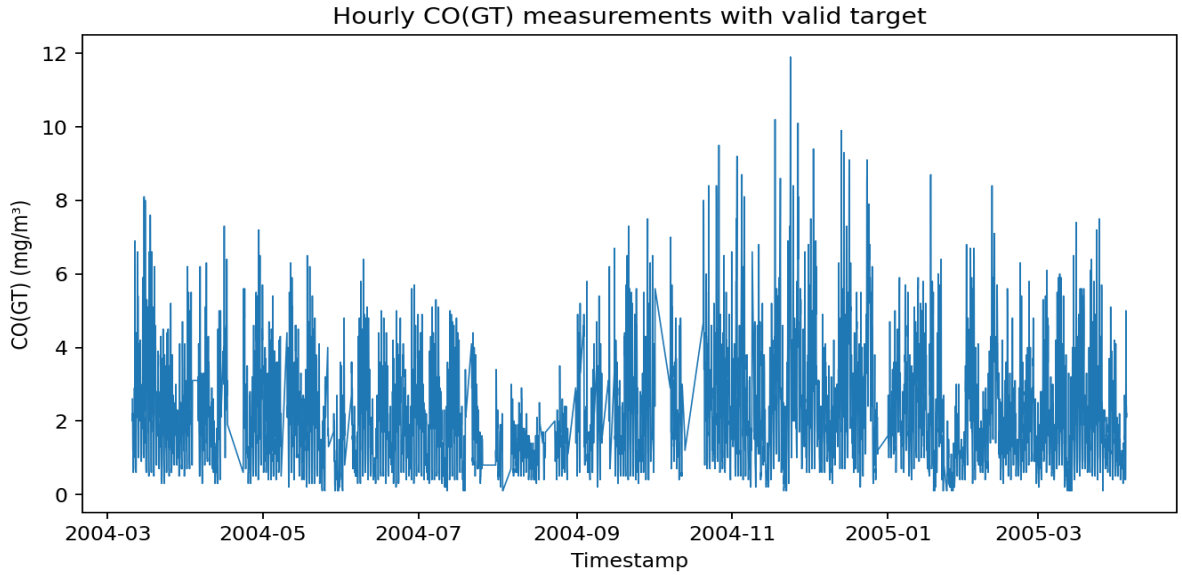


Figure 1. Valid hourly CO measurements over time.

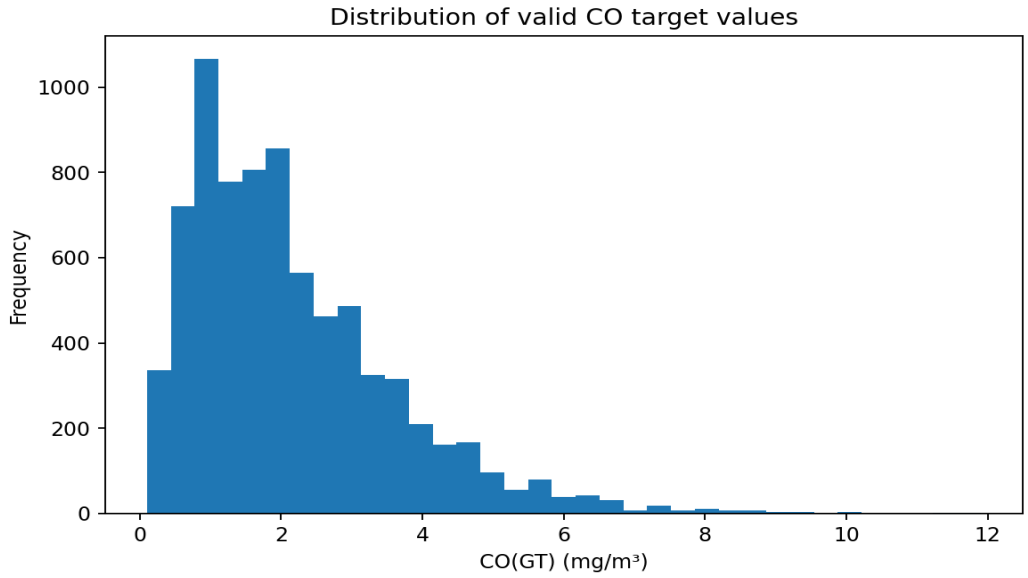


Figure 2. Distribution of valid CO target values.

6. Preliminary Model Comparison

Reference benchmark from analogous Python model-family implementations, included as a sanity-check only because R was unavailable during package assembly. The R script is authoritative; replace these values with outputs/test_metrics.csv after running in RStudio.

Model	MAE	RMSE	R^2
Random Forest Regression	0.318	0.480	0.858
Gradient Boosting Regression	0.321	0.481	0.858
Support Vector Regression (linear kernel)	0.336	0.495	0.849
Multiple Linear Regression	0.372	0.515	0.837
KNN Regression	0.390	0.550	0.814

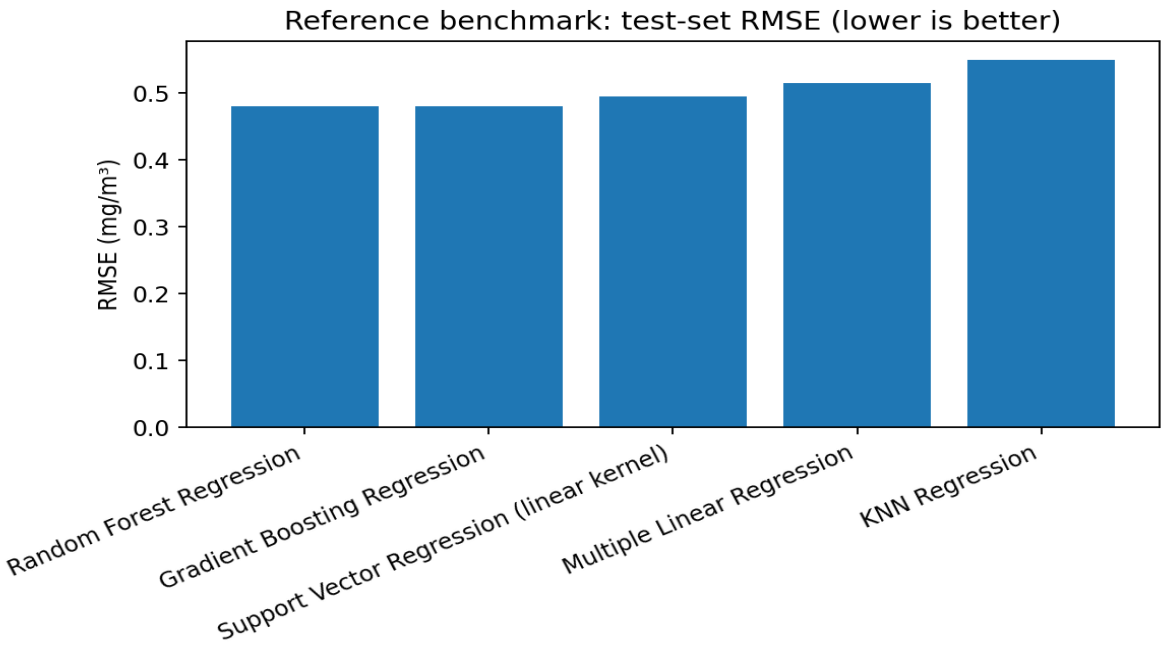


Figure 3. Reference benchmark test RMSE; lower is better.

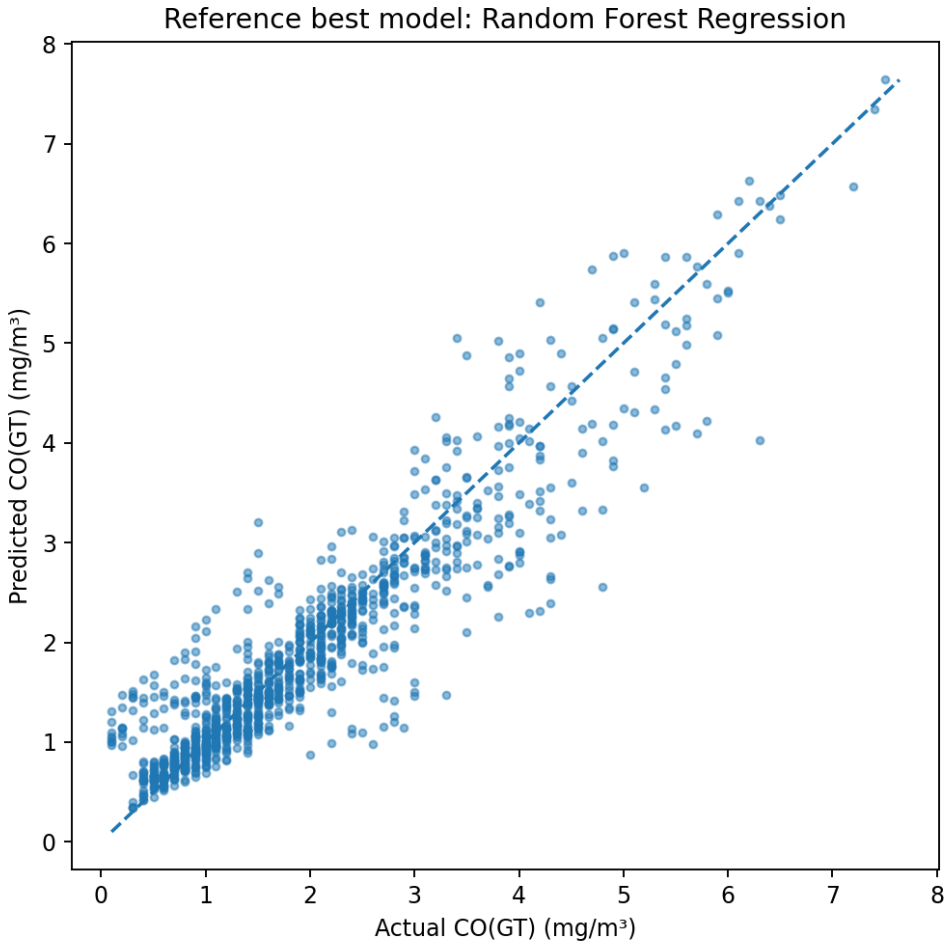


Figure 4. Actual versus predicted reference plot for Random Forest Regression.

7. High-Pollution Error Audit

The R pipeline writes outputs/high_pollution_error_audit.csv, comparing overall test error with errors among observations at or above the training-plus-validation CO 75th percentile. Review this output after running the R script.

8. Limitations and Future Work

- One location and historical observations (2004–2005); limited external validity.
- Sensor drift and missingness can affect estimates.
- Same-hour estimation, not future forecasting.
- The benchmark scores are not the final R output; regenerate them using the included R script.

Future work: one-hour-ahead forecasting with lagged features, rolling-origin validation, nonlinear SVR, and external validation on a second dataset.

References and Reproducibility

Vito, S. (2008). Air Quality. UCI Machine Learning Repository. https://doi.org/10.24432/C59K5F
R Core Team. https://www.r-project.org/

Run from the project root: source('scripts/run_project.R'). Outputs are saved under outputs/.